

Oil spill detection using GLCM and MRF

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Abstract—This paper presents a study for oil spill detection in three steps. The first one considers the texture as a two dimensions array, and to describe the statistics iteration between pixels the algorithm computes a textural feature related with the Gray Level Co-occurrence Matrix (GLCM). After, the original image and the textural feature images are segmented using Markov Random Field (MRF). Each pixel can be classified in two classes: {oil, not-oil}. To determine the class we optimized the *a posteriori* energy function by means of simulated annealing. The segmentation result contains different levels of information, in order to improve the oil spill detection; we propose a data fusion stage. The result obtained is binary and shows in detail the oil spill in the analysis zone.

Keywords: MRF, GLCM, SAR, Segmentation, oil spill.

I. INTRODUCTION

The Remote Sensing is the technology to acquire data and information about an object or phenomena by a device that is not in physical contact with it. In other words, remote sensing refers to get information about the earth and its environment by distance.

Most common mediums include satellite imaging, GPS, Radar, Sonar and aerial photography. One of the main instruments used in remote sensing is the Synthetic Aperture Radar (SAR). This instrument presents the advantage to provide its own illumination source, allowing the data acquisition under all climatologic conditions. Its disadvantage consists in the presence of *speckle* noise, very common in systems that use sources of coherent illumination.

One emerging SAR application is oil spill detection. There are different representative frameworks to oil spill detection like [1] using statistics fundamentals as minimum description length principle or a set of algorithms to classify the dark spot in [2].

This algorithm to oil spill segmentation in SAR images, using Gray Level Co-occurrence Matrix, segmentation by mean of Markov Random Fields and image fusion.

The proposed algorithm is applied in a SAR image acquired by the ERS-2 satellite, with a pixel size of 12.5 m x 12.5 m. The image belongs of the Prestige oil spill disaster along Galicia's coast in November 2002.

II. GRAY LEVEL CO-OCURRENCE MATRIX

The SAR images are very rich in texture. The common framework to handle these kinds of images is applying a pre-filter stage but that means loose information. The Gray Level Co-occurrence Matrix (GLCM) was used to texture modelling without loose special information. The definition of GLCM is as follows [3]. Suppose an image to be analyzed is rectangular and has N_x columns and N_y rows. Suppose that the gray level appearing at each pixel is quantized to N_g levels. Let $L_x = \{1, 2, \dots, N_x\}$ be the columns, and $L_y = \{1, 2, \dots, N_y\}$ be the rows, and $G_x = \{0, 1, 2, \dots, N_g - 1\}$ be the set of N_g quantized gray levels. The set $L_y \times L_x$ is the set of pixels of the image ordered by their row-column designations. The image I can be represented as a function that assigns some gray level in G to each pixel or pair of coordinates in $L_y \times L_x$; $I: L_y \times L_x \rightarrow G$.

The texture information is specified by the matrix of relative frequencies P_{ij} with two neighboring pixels separated by a distance δ in the calculation window, one with gray level i and the other with gray level j .

Such matrices of gray level co-occurrence frequencies are a function of the angular relationship and distance between the pixels. Formally, in this analysis, for angles of 0° and 180° , the unnormalized frequencies is defined as shown in the next equation. The # denotes the number of elements in the set.

$$P(i, j, \delta) = \# \{ ((k, l), (m, n)) \in (L_y \times L_x) \times (L_y \times L_x) \mid k - m = \delta, l - n = 0 \} \quad (1)$$

and for 90° and 270° the equation is:

$$P(i, j, \delta) = \# \{ ((k, l), (m, n)) \in (L_y \times L_x) \times (L_y \times L_x) \mid k - m = 0, |l - n| = \delta \} \quad (2)$$

In designing the GLCM for texture representation, there are three fundamental parameters: quantization levels, displacement and orientation values of the measurements.

The orientation parameter is less important compared to other factors in co-occurrence matrices. For SAR oil spill images, there are no systematic patterns based on orientation. The oil spill can be in all possible orientations. Then, we argue that the orientation factor is not important.

A quantization scheme is needed in order to perform the GLCM computation. The uniform quantization was chosen as the simplest. The gray level selection is not trivial and then without loss of generality if the image presents k different number of classes then the number of gray levels will be $k+1$.

The displacement parameter δ is important in the computation of GLCM. Applying a large displacement value to a fine texture, the Co-occurrence Matrix will not capture the textural information. The displacement value selected was equal to the finest details present in the image.

We consider that all textural information is contents in the GLCM and the textural features will be applied over it to describe the image; the textural feature used in this work was correlation:

$$f_3 = \sum_i \sum_j \frac{(i - \mu_x)(j - \mu_y)}{\sigma_x \sigma_y} \cdot p(i, j) \quad (1)$$

where:

$$\mu_x = \sum_i \sum_j i \cdot p(i, j) \quad (2)$$

$$\mu_y = \sum_j \sum_i j \cdot p(i, j) \quad (3)$$

$$\sigma_x^2 = \sum_i \sum_j (i - \mu_x)^2 \cdot p(i, j) \quad (4)$$

$$\sigma_y^2 = \sum_j \sum_i (j - \mu_y)^2 \cdot p(i, j) \quad (5)$$

III. MARKOV RANDOM FIELD SEGMENTATION

Using segmentation methods based on Markov Random Fields (MRF) need a huge computing power and a lot of time. For these reasons, MRF methods are used in offline tasks, never in a real time processing environment.

By definition the MRF has the property that the conditional distribution of a particular pixel given the values of all the other pixels in the whole image is equal to the conditional distribution given only by the pixel values in its neighborhood.

This property is important and, when dealing with a particular pixel, we take in count all those pixels inside its neighborhood; this is the main reason of why the Markov Random Fields incorporate a great capability in noisy images segmentation [4].

If Y is the segmented image, which is started randomly with uniform probability density function, and X is the SAR image, then the possibility to belong to a particular class is given minimizing the *a posteriori* energy function $U(Y|X)$.

The *a posteriori* energy function is defined as follows:

$$U(Y|X) \approx U(X|Y) + U(Y) \quad (6)$$

$U(Y)$ is the *a priori* energy function and $U(X|Y)$ is the joint energy function.

The algorithm realizes the segmentation under two different *a priori* energy functions approaches: in 8-conexity and 4-conexity.

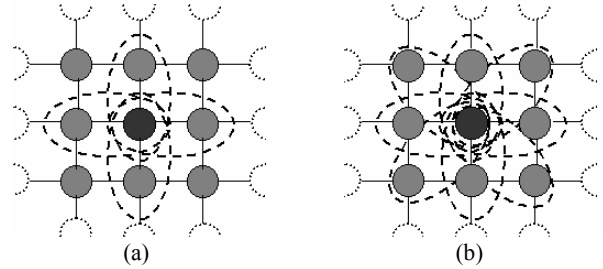


Figure 1. Conexity analysis. (a) 4-conexity. (b) 8-conexity.

The *a priori* function energy $U(Y)$ is defined as:

$$U(Y) = -\beta \times n_i(Y) \quad (7)$$

Where $n_i(Y)$ is the number of pixels in the neighborhood with label Y and β is a homogeneity coefficient.

To compute the joint energy function we define training windows over the SAR images to find the mean $\{\mu_{oil}, \mu_{not-oil}\}$ and we approach the standard deviation s as we show in Fig. 2.

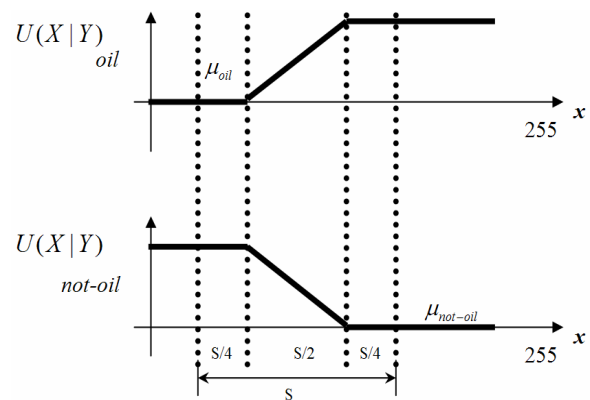


Figure 2. Joint energy function.

Once the global energy has been defined, a variety of algorithms can be used to find the global minimum.

Traditional approaches associated with regularization and Markov Random Fields include simulated annealing [4], or

mean field annealing [5]. More recently, max flow and graph cut methods have been proposed to solve global optimization problems in image segmentation [6]. The global energy function was optimized by means of simulated annealing.

The algorithm segmented the correlation image (GLCM) and the original SAR image.

IV. IMAGE FUSION

In this analysis we made use of the original SAR image and the correlation image, both segmented by mean of MRF 8-conexity scheme.

The image fusion produces a single image from a set of input images. The fused image should have more complete information than the original ones.

The result contains two kinds of information: complementary information and redundant information.

To obtain and improve the quality of the final result, first we calculate the error measures. Some raw information introduced via handmade map is also taken in count.

Taking x as an image pixel, $L=\{oil, not-oil\}$ denote the set of classes. If $M(x)=l_i$ is the class of the pixel x according to the map, and $I(x)=l_j$ is the class of pixel x given by an image analysis result [7].

Let x' be a pixel in the neighborhood V_x of pixel x , and $d(x, x')$ the euclidean distance between x and x' .

We calculate the following two measures:

$$a(x) = \sum_{x' \in V_x} \frac{1\{M(x')=l_j\}}{d(x, x')} \quad (8)$$

$$b(x) = \sum_{x' \in V_x} \frac{1\{I(x')=l_j\}}{d(x, x')} \quad (9)$$

Where $a(x)$ is the number of pixels in V_x which are classified as l_j in the map, inversely weighted by $d(x, x')$.

The Error measure is calculated as follows:

$$E(x) = \frac{1}{a(x)+b(x)+1} \quad (10)$$

Here $0 < E(x) \leq 1$, but if $l_j = l_i$ then $E(x) = 0$.

The reliability is calculated based on the local error measure of the site:

$$C(x) = 1 - E(x) \quad (11)$$

The final decision is made by taking the reliability coefficient $C(x)$ to calculate the accumulation of the same site.

$$l(x) = \max_{l_i \in L} \sum_{k=1} C_k(x) 1_{\{I_k(x)=l_i\}} \quad (12)$$

Where $1_k(x)$ means the result used as the K_{th} input to the fusion process, and $C_k(x)$ the confidence coefficient of pixel x in this result.

V. RESULTS

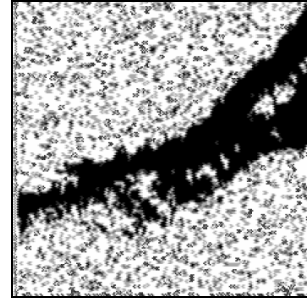


Figure 3. Correlation image.

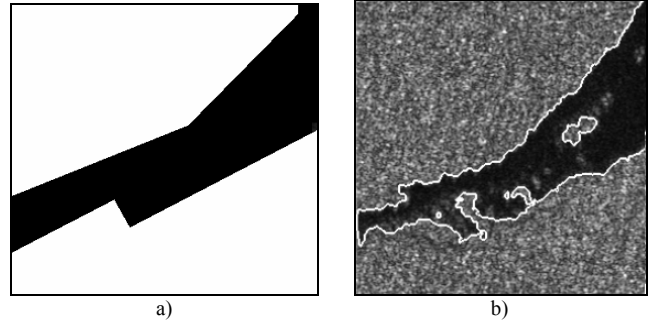


Figure 4. a)Raw map , b)SAR original image MRF segmentation

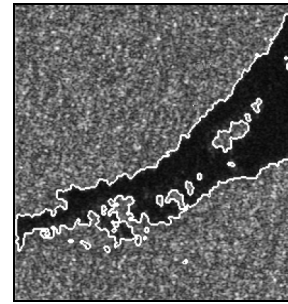


Figure 5. Result

Figure 4.a) shows a raw segmentation proposed to the data fusion algorithm. Figure 4.b) shows the original image under a typical MRF segmentation. Figure 5 shows the final result of the fused images.

VI. CONCLUSIONS

SAR segmentation is a hard task because the *speckle* noise is present. One way to solve this problem, without loss information, is using a contextual analysis using MRF theory.

The first stage of the proposed scheme was the segmentation of the original SAR image and the segmentation of the correlation image related with the GLCM.

The *a priori* energy function $U(Y)$ was associated with the image that is initialized randomly in the simulated annealing. The analysis of the neighborhood was restricted for 8-conexity.

We realize the image fusion for the two results and introduce a raw map to the fusion algorithm. The process increases the quality of the final results. Next work will be devoted to modelling the gray tone spatial dependence of sea regions by statistical analysis using only Gray Level Co-occurrence Matrix reducing the time processing.

REFERENCES

- [1] F. Galland, P. Refregier and O. Germain, "Synthetic Aperture Radar oil spill segmentation by stochastic complexity minimization", IEEE Geoscience and Remote Sensing Letters, Volume: 1, Issue: 4, Pages: 295 – 299, Oct. 2004.
- [2] Solberg, G. Storvik, R. Solberg, and E., "Automatic detection of oil spills in ERS SAR images", IEEE Transactions on Geoscience and Remote Sensing, Volume: 37, Issue: 4, pp. 1916 – 1924, July 1999.
- [3] R. M. Haralick, K. Shanmugam, and I.Dinstein, "Texture features for image classification", IEEE Trans. Syst. Man Cybern., Volume: SMC-3, no. 6, 610-62, 1973.
- [4] S. Geman and D. Geman, "Stochastic relaxation, Gibbs distribution, and the Bayesian restoration of images", IEEE TPAMI, Volume 6, Issue 6, pp. 721–741, 1984.
- [5] D. Geiger and F. Girosi. "Parallel and deterministic algorithms for MRF's: Surface reconstruction", IEEE TPAMI, Volume 13, Issue 5, pp. 401–412, 1991.
- [6] Y. Boykov, O. Veksler, and R. Zabih, "Fast approximate energy minimization via graph cuts" IEEE TPAMI, Volume 23, Issue 11, pp.1222–1239, 2001.
- [7] S. Yu, "Improving Satellite Image Analysis Quality by Data Fusion", IEEE IGARSS, pp. 2164-2166, 1995.